JAVA BE INTERVIEW – TOPIC WISE MASTER

Developed by dhruvtechbytes

@Instagram - https://www.instagram.com/dhruvtechbytes

Java/Jre/JVM

1. What is Java Virtual Machine (JVM)? Difference between JVM/JDK /JRE?
2. Explain the JVM architecture and its main components (Class Loader, Runtime Data Areas, Execution Engine). What are the different runtime data areas in JVM (Heap, Stack, Metaspace, Code Cache)?
3. **What are the thread-specific runtime data areas in JVM? Explain their roles.**
   Or
   What is the impact of thread creation on memory allocation in these areas?
4. Explain the Java Memory Model (JMM) and its guarantees.
5. What are happens-before relationships in JMM?
6. How does the JVM implement volatile variables and atomic operations?
7. Your production service shows occasional spikes in response time. GC logs indicate frequent Full GCs. How would you analyse and check the causes and provide resolution?
   Or
   How do you analyze and tune JVM performance using JFR, JIT logs, and GC logs?
8. Various JVM or java args that can help fine tune the java and JVM performance?
9. What is MetaSpace? How does it differ from PermGen?
10.

Java 8:

1. What are Functional interfaces? Why they were added in Java 8?Is FI annotation mandatory?
2. Various types or categories of FI?Any predefined Fis(ex. Runnable)? What is the difference between Predicate, Function, and Supplier interfaces?
3. Write a FI for basic tasks like summing 2 numbers or any FizzBuzz and use it.
4. What is the default method, and why is it required?
5. Can a functional interface extend/inherit another interface? A functional interface cannot extend another interface with abstract methods as it will void the rule of one abstract method per functional interface. It can extend other interfaces which do not have any abstract method and only have the default, static, another class is overridden, and normal methods.
6. Method references? Syntax? Scenario based question to check if he knows the syntax well as well as existing method references added in library: You have a list

of strings and want to sort them ignoring case. How can you use method references instead of lambdas?

7. What is the difference between lambdas and anonymous inner classes in terms of scope and performance?
8. Asynchronous Programming? CompletableFuture?
9. Diff between terminal and intermediate operations in Java 8? Test ability to identify in real programs
10. Parallel streams? When to use? What are the pitfalls of using parallelStream() in a web application?
11. What is a Spliterator in Java 8?Diff between Iterator vs Splititerator?
12. Is multiple inheritance possible in java 8? If not why? What about Multi level inheritance?Diamond prob?

Collections

1. Internan working of Arraylist{should know when n how new array is created and data copied}? HashMap{should know hashing and then index generation, data stored in buckets in array and hash Collisions lead to linked list[singly] creation,RB tree when? need of Hashcode, equals, Immutable classes}? What happens when you put a key object in a HashMap that is already present?
2. Why Concurrenthashmap when already HashTable present? Segmentation? Locking? Concurrent modification?
3. What are the differences between the two data structures: a Vector and an ArrayList?
4. **In which scenario, LinkedList is better than ArrayList in Java?**
5. **What are the differences between a HashMap and a Hashtable in Java?**
6. **Internal working of treemap? Diff between Treemap and hashMap?**
7. **Immutable class? Why are these needed? How to make them secure from I/o and reflection?examples**
8. **Shallow and Deep cloning? When to use which? Which one is needed while creating Immutable class?**
9. **Comparator vs Comparable? Which one provides natural sorting? Implement same for employee[id, name. salary,age] to sort in descending order of salary?**
10. **Collections class? When to use synchronized collections? Arrays class?**
11. **What is CopyOnWriteArrayList? How it is different from ArrayList in Java?**
12. **WeakHashMap? WeakReference? SoftReference?**
13. **Let say there is a Customer class. We add objects of Customer class**
14. **to an ArrayList. How can we sort the Customer objects in ArrayList by**
15. **using customer firstName attribute of Customer class?**

16. What is the difference between Synchronized Collection and Concurrent Collection?
17. What is the scenario to use ConcurrentHashMap in Java?
18. How will you create an empty Map in Java?
19. What is the difference between remove() method of Collection and remove() method of Iterator?
20. Between an Array and ArrayList, which one is the preferred collection for storing objects?
21. Is it possible to replace Hashtable with ConcurrentHashMap in Java?
22. How CopyOnWriteArrayList class is different from ArrayList and Vector classes?
23. Why ListIterator has add() method but Iterator does not have?
24. Why do we sometime get ConcurrentModificationException during

iteration?

25. How will you convert a Map to a List in Java?
26. How can we create a Map with reverse view and lookup in Java?
27. How will you create a shallow copy of a Map?
28. Why we cannot create a generic array in Java?
29. What is a PriorityQueue in Java?

Exception handling

1. What is the base class for Error and Exception classes in Java?
2. What is a finally block in Java? Output of following

```
public class FinallyTest {
  public static int testMethod() {
    try {
      System.out.println("In try block");
      return 10;
    } catch (Exception e) {
      System.out.println("In catch block");
      return 20;
    } finally {
      System.out.println("In finally block");
      return 30;
    }
  }
```

```
    public static void main(String[] args) {
      System.out.println("Result: " + testMethod());
    }
  }
```

3. In Java, what are the differences between a Checked and Unchecked? Examples? How would you handle these?
4. Can we create a finally block without creating a catch block? In what scenarios, a finally block will not be executed?
5. What is the concept of Exception Propagation?Example by code?
6. When we override a method in a Child class, can we throw an additional Exception that is not thrown by the Parent class method?
7. Try with resource? Any changes done in it recently? When to use?

String

1. What is the meaning of Immutable in the context of String class in Java?
2.Why a String object is considered immutable in java?
3.How many objects does following code create?
4.How many ways are there in Java to create a String object?
5.How many objects does following code create?
6.What is String interning?
7.Why Java uses String literal concept?
8.What is the basic difference between a String and StringBuffer object?
9.How will you create an immutable class in Java?
10.What is the use of toString() method in java ?
11. Arrange the three classes String, StringBuffer and StringBuilder in the order of efficiency for String processing operations?
30. Diff between or how many objects are created when
    String s=new String(“Shiv”);
    Struing s1=”Shiv”;
    String s2=”Shiv”;
31. Concept of interning? Output of below program

```
 String s1 = "Java";
String s2 = "Java";
String s3 = new String("Java");
String s4 = s3.intern();
System.out.println(s1 == s2);
```

System.out.println(s1 == s3);
System.out.println(s1 == s4);

32. **Output of below:**

```
String s = "Hello";
s.concat("World");
System.out.println(s);
```

33. **How many objects**

```
String s1 = "abc";

String s2 = new String("abc");

String s3 = s2.toUpperCase();
```

2

Multithreading

1. **Thread vs Runnable? Creating threads in diff ways?Runnable as lambda?**
2. **Different ways of synchronization of threads in Java?**
3. **What does the synchronized keyword do in Java? How does it work internally?**
4. **What is the difference between synchronizing on a method vs synchronizing on a block?**
5. **What is the difference between synchronized instance methods and synchronized static methods?**
6. **What object is used as the lock when you declare a method as synchronized?**
7. **Can constructors be synchronized? Why or why not?**
8. **What happens if two threads call two different synchronized methods on the same object?**
9. **What is the difference between synchronized(this) and synchronized(SomeClass.class)?**
10. **What is the impact of synchronized on performance compared to ReentrantLock?**
11. **How does JVM implement synchronization under the hood (monitor enter/exit, biased locking)?**
12. **What happens if a thread holding a lock calls wait()? Does it release the lock?**
13. **Race Condition vs Visibility problem? How volatile solves visibility?**
14. **JMM in detail? Happens before?**
15. **What is a daemon thread in Java?How will you make a user thread into daemon thread if it has already started?**
16. **Wait(), notify(), notifyAll()? Present in which class and why?**
17. **Runnable vs Callable?**

18. ExecutorService? Type of thread pool? What considerations while creating thread pool?
19. What is the fundamental difference between wait() and sleep() methods?
20. Locks? What is the difference between synchronized and Lock interface in Java? Wh-
21. Explain the difference between ReentrantLock and intrinsic locks (synchronized).
    (Explicit lock/unlock, fairness policy, condition variables.)
22. What is a reentrant lock? Why is it called reentrant? Main purpose of using them? Fairness?diff with ReentrantReadWritelock?
23. What is the use of ThreadLocal class?
24. There are two threads T1 and T2? How will you ensure that these threads run in sequence T1, T2 in Java?
25. What is difference between CyclicBarrier and CountDownLatch class?

Miscellaneous:

1. Serializable ? Significance of serialVersionUID? Externalizable?
2. Marker interfaces? Why required?
3. Is there any difference between a = a + b and a += b expressions?
4. Let say there is a method that throws NullPointerException in the superclass. Can we override it with a method that throws RuntimeException?
5. Can we make an array volatile? Any caveats? How can you mark an array volatile in Java properly?
6. Which class contains clone method? Cloneable or Object class? Is Cloneable broken interface?
7. What will this return 5*0.1 == 0.5? true or false?

Spring/Spring boot:

1. IOC? Purpose and benefits of IOC container in Spring?DI? types?
2. Singleton in Spring vs Singleton implementation in java? Is it safe to assume that a Singleton bean is thread safe in Spring Framework?
3. Lifecycle of a bean?
4. Autowiring? Different methods of autowiring? What if we implement same interface in two classes and we have to inject one of the class using autowiring? @Primary, @Qualifier, Convention over configuration in qualifying by object name?
5. @Configuration? @Bean?
6. @Autowired vs @Inject

7. **Diff between ResponseBody and ResponseEntity?**
8. **Diff between Controller and RestController? Why @ResponseBody not needed in @RestController**
9. **Significance of @SpringBootApplication annotation? What annotations does it comprise of and function of each of them?**
10. **How does @ComponentScan work if no package name given with annotation?**
11. **How does @EnableAutoConfiguration in Spring Boot enable plug-and-play functionality?**

Spring Controller and Web layer:

1. **How do you handle JSON and XML responses in Spring Boot REST APIs?**
2. **What is the difference between @RequestParam, @PathVariable, and @RequestBody? Give implementation for each and share how you will generate endpoint and pass values**
3. **Best practices while creating endpoints, url naming scheme?**
4. **Various REST methods like GET,PUT,POST.PATCH.OPTIONS, DELETE? Which are idempotent?**
5. **How do you implement file upload and download in Spring REST?**
6. **How do you configure CORS in Spring Boot REST APIs?**
7. **How does Spring Boot auto-configure Jackson for JSON serialization/deserialization?**
8. **Explain the lifecycle of a REST request in Spring Boot (DispatcherServlet -> HandlerMapping -> Controller -> ViewResolver).**
9. **Richardson maturity Model and hateos? How do you implement HATEOAS in Spring Boot REST?**
10. **How do you secure REST APIs using Spring Security + JWT? JWT structure? JWT lifecycle?**
11. **Rate limiting, Throttling in Spring boot application? What steps are required?**
12. **How to handle caching with Spring boot? @EnableCaching ? How does @Cacheable work in Spring Boot? What is the difference between @Cacheable, @CachePut, and @CacheEvict?Cache expiration handling?**
13. **How do you implement conditional caching using unless and condition attributes in @Cacheable?**
14. **Caching in clustered env? How do you prevent cache stampede or cache avalanche in Spring Boot?**
15. **How do you implement ETags and conditional GET with caching for REST APIs?**

16. How does @Cacheable works internally? CacheInterceptor.Cache Manager.. Cache resolver? Cache lookup...decision baed on presence...
17. How do you implement multi-level caching (in-memory + Redis)?When you will use it?Disads?
18. Dispatcher servlet? Role?Explain the lifecycle of DispatcherServlet from initialization to request handling. How does DispatcherServlet handle asynchronous requests (DeferredResult, Callable)?
19. How does DispatcherServlet interact with HandlerInterceptor and what is the exact order of execution?
20. How do you validate request data in a Spring Boot controller? @Valid and @Validated?Diff bw two?
21. How do you implement custom validation annotations in Spring?
22. How do you validate nested objects in a request payload? How do you validate collections (e.g., list of objects) in a request body?
23. How do you handle validation for path variables and query parameters?
24. How do you implement cross-field validation (e.g., password and confirm password)?
25. How do you customize the MethodValidationPostProcessor for advanced use cases?
26. How do you validate file uploads or multipart requests in Spring?

Exception handling:

1. How do you handle exceptions in Spring Boot using @ExceptionHandler?
2. What is the difference between @ControllerAdvice and @RestControllerAdvice?
3. Create and Handle custom exceptions
4. How do you handle validation exceptions (MethodArgumentNotValidException)?
5. How do you propagate exceptions from service layer to controller layer?
6. How do you use ResponseEntityExceptionHandler for centralized exception handling?

JPA/JDBC

1. What is diff between JPA, Hibernate/Ibtis, Spring Data JPA?
2. Use of transient keyword?
3. Create an Entity for Employee[int id, String name, int age, double salary] with all annotations
4. What is the purpose of @GeneratedValue in JPA? What are the different strategies supported by @GeneratedValue? Explain the difference between GenerationType.AUTO, IDENTITY, SEQUENCE, and TABLE.

5. How does GenerationType.AUTO decide which strategy to use?
6. When would you use GenerationType.SEQUENCE over IDENTITY?perf implications?
7. What is the role of @SequenceGenerator and @TableGenerator?

```
@GeneratedValue(strategy = GenerationType.TABLE, generator = "prod_gen")
@TableGenerator(name = "prod_gen", table = "id_generator", pkColumnName = "gen_name",
        valueColumnName = "gen_value", allocationSize = 5)

@GeneratedValue(strategy = GenerationType.SEQUENCE, generator = "emp_seq")
@SequenceGenerator(name = "emp_seq", sequenceName = "employee_sequence", allocationSize = 10)
```

8. How do you implement a custom ID generator in JPA?
9. How does JpaRepository differ from CrudRepository?
10. How do you use @Query annotation for custom JPQL queries?
11. How do you enable pagination and sorting in Spring Data JPA?
12. What is the role of EntityManager in Spring Data JPA?
13. When to use @OneToOne, @OneToMany, @ManyToOne, @ManytoMany?
14. Explain the difference between fetch = FetchType.LAZY and FetchType.EAGER.
15. N+1 query problem? caused by FetchType.EAGER or lazy loading?
16. How do you use @Query annotation for custom JPQL queries? Let's say we want to modify data in Table using @Query what other annotations required?
17. Use of @modifying and @Transactional?
18. How do you implement auditing in Spring Data JPA (@CreatedDate, @LastModifiedDate)?
19. How do you handle optimistic and pessimistic locking in Spring Data JPA?
20. @version? @Lock? @Transactional?various lock modes?
21. How do you integrate Spring Data JPA with multiple data sources?
22. How do you implement multi-tenancy with Spring Data JPA?application connects or holds data for multiple orgs or clients....How to handle this in Spring Boot app? DB per tenant, Schema per tenant, shared schema discriminator based
23. How does Spring Data JPA work with native queries and projections?
24. How do you debug and optimize slow queries in Spring Data JPA?
25. How do you handle bidirectional relationships and avoid infinite recursion in JSON serialization?
26. How do you implement cascade operations in JPA?

27. How do you use Entity Graphs for optimizing queries?
28. How do you implement batch inserts/updates efficiently?
29. How do you define a Named Query using @NamedQuery annotation? What is the difference between @NamedQuery and @NamedNativeQuery?
30. How does Spring Data JPA resolve Named Queries internally?
31. What is the precedence between Named Queries and @Query annotation in Spring Data JPA? How do you override a Named Query defined in an entity?
32. How do you define an index in JPA using annotations?
33. What is the difference between @Index and unique=true on @Column?
34. Advantage and disadvantages of indexing? What is the impact of indexing on insert/update performance?
35. Indexes in joins? Does JPA automatically create indexes for foreign keys?

DB

1. DDl,DML? Diff between Delete and truncate? WHERE vs HAVING?
2. Explain INNER JOIN vs LEFT JOIN with examples.
3. ACID properties?
4. Indexing? When to do? Advantage /Disadvantages
5. How do you implement pagination in SQL?
6. Views over tables? Why use views?View vs Materialized Views? Ad/Disad?
7. Arrange in orderof execution
   WHERE,GROUP BY,SELECT ,HAVING,FROM,JOIN ,ORDER BY ,LIMIT / OFFSET

   Correct: FROM, JOIN, WHERE, GROUP BY, HAVING, SELECT, ORDER BY, LIMIT / OFFSET
8. Self join? Why it is required?]
9. Find name of Employee having second or n max salary. Emplouyee[id, name,salary,dept]
10. find employees who earn more than the average salary of their department, and rank them within their department by salary.
11. Find the top 3 highest-paid employees in each department.
12. Given Employee(id INT, name VARCHAR(50), department_id INT, manager_id INT, salary DECIMAL, hire_date DATE)
13. Find employees who earn more than the average salary of their department.
14. Find employees who joined before the earliest hire date in their department.
15. Find employees who earn more than the average salary of their department AND more than their manager.
16. Find employees who have the same salary as someone in another department.

17. **Partitioning?Different types of partitioning? (Range, List, Hash)**
18. **What is the difference between horizontal partitioning and vertical partitioning?**
19. **How do you manage indexes on partitioned tables?**
20. **What is sub-partitioning and when would you use it?**
21. **How do you implement partitioning with composite keys?**
22. **Composite keys?Primary keys? Candidate Keys? Surrogate Keys? Is it possible to have a table without a primary key in a relational database? If yes, what are the implications for data integrity and query performance? How would you uniquely identify rows in such a table?**

Microservices

1. **Microservices in your own words? How is it diff from Monolith?benefits of using MS and Downsides?**
2. **How do microservices communicate with each other? Synchronous and Asynchronous modes? REST/HTTPS, SOAP, gRPC, Message queues like RabbitMQ,KafKa etc.**
3. **Explain Event Driven Architecture? How,Why and when used?**
4. **Explain the difference between an event, a command, and a query in EDA.**
5. **What is the role of an Event Bus or Message Broker in EDA?**
6. **Explain the concept of Event Sourcing and how it differs from traditional CRUD.**
7. **What are the challenges of implementing distributed transactions in EDA and how do you solve them? Using Saga pattern how can we solve these problems?**
8. **Saga pattern?Different types? When how and why use? What happens if one step in a Saga fails?How does it ensure eventual consistency?**
9. **How do you implement compensating transactions in a Saga? How do you handle retries and idempotency in Saga steps?**
10. **What are the challenges of implementing Saga in a system with high throughput?How about when need to design a Saga for a globally distributed system with strict latency requirements?**
11. **CQRS?Differentiate Command and Query ? benefits? Disads? How do you keep the read model in sync with the write model?**
12. **Explain how eventual consistency works in CQRS.**
13. **How to implement feign client in an application? What are various steps? Where do you manage URls?How do you handle fallbacks?**
14. **difference between fault tolerance and resilience.**
15. **How to ensure resilience in MS?What are various resilience patterns?**

Circuit Breaker? Bulkhead?

16. Load balancing? client-side and server-side load balancing in microservices? Explain any load balancing scheme used in any of your project?
17. How does Kubernetes Service load balancing differ from an API Gateway load balancer?
18. What is the difference between Layer 4 and Layer 7 load balancing? Which one is better for microservices?
19. Steps to implement a Circuit breaker library [Resilience4j or Hysterix]? various states handling?
20. How can retry,retry with exponential backoffs, timeouts and rate limiting help? How to implement them
21. Scalability? Horizontal vs Vertical? Are MS more scalable than monolith?why?
22. How do you implement auto-scaling in a Kubernetes-based microservices architecture?HPA, VPA,CA etc
23. 12 factor apps standards? CAP theorem?
24. What is service discovery and why do we need it?Steps to implement the same in Spring boot?
25. What is distributed tracing and why is it important in microservices?concept of traceID n span Id?How do you pass or propagate trace context though all the microservices?Implementing it in spring boot? Libraries to be used?

© dhruvtechbytes | Java • Spring Boot • Microservices • Database